

Kratik Rath

+1 (812) 822-7764 | rathi.kratik08@gmail.com | [linkedin.com/kratikrathi](https://www.linkedin.com/in/kratikrathi) | [Kratik-Rathi-Portfolio](#) | github.com/Kratik-Rathi

EDUCATION

Indiana University - Bloomington

Master of Science in Data Science

Coursework: Information Visualization, Applied Database Technologies, Applied Machine Learning, Intro to Statistics, Management Access of Big Data

August 2024 – May 2026

GPA: 3.6

Medi-Caps University

Bachelor of Technology in Computer Science and Engineering

August 2019 – July 2023

GPA: 3.4

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, C++, C#

Databases & Warehousing: Snowflake, BigQuery, MySQL, PostgreSQL, MS SQL Server, MongoDB

Cloud Platforms & Big Data Technologies: Databricks, Apache Spark (PySpark), AWS, GCP

ETL & Data Visualization: Alteryx, Power BI (DAX, Power Query), Tableau, Looker Studio, Excel

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Keras, OpenCV, LangChain, Matplotlib, Seaborn, Streamlit, Flask, .NET

Machine Learning: Regression Models, Tree-based Models (XGBoost, Random Forest), SVM, Clustering (K-Means, GMM), Deep Learning (CNN)

Development Tools: Git (GitHub, GitLab), Docker, Postman

WORK EXPERIENCE

Data Analyst at Project 990 | Bloomington, IN

October 2025 – Present

- Processed and analyzed 1M IRS tax-return records in Alteryx to build a nonprofit network and generate grant-making / grant-seeking recommendations.
- Built a Tableau geographic map of 400K U.S. nonprofits to help grant makers identify and target high-need areas.

Software Engineer Intern at Genmark AI Inc. | Remote, USA

July 2025 – August 2025

- Developed the initial version of Gemini-powered RAG pipeline using Python and LangChain, embedding crawled data from GCP Cloud Bucket into a Milvus vector database and containerizing the end-to-end system with Docker to support scalable, context-aware retrieval.
- Enabled approximately 60% faster content generation for clients by implementing a chat-based interface over embedded assets, producing accurate context-aware responses using Gemini and a Dockerized vector store.
- Reduced content creation time to under 5 minutes by building a tool that analyzed top-20 trending TikTok videos, generated transcripts via Whisper and produced personalized videos using Google Veo2; recognized as "Most Innovative Project".

Research Assistant at Indiana University – Indiana Innocence Project | Bloomington, IN

January 2025 – May 2025

- Constructed a Python-based ingestion pipeline using BeautifulSoup to scrape and normalize legal case data from intake forms and judicial sources into a PostgreSQL database, cutting manual data entry time from 30 minutes to 1 minute.
- Designed a Looker Studio dashboard to visualize case timelines, severity and procedural status, helping assign review priority to 200+ cases.

Data Analyst Intern at Swastika Investmart Ltd. | Indore, India

January 2024 – July 2024

- Created an automated analytics workflow using Python, Flask API, SQL and cron jobs to generate daily reports on 50K+ records, providing product and campaign level insights across client zones, which led to a 20% increase in conversions and reduced manual effort by 90%.
- Engineered a scalable Docker-based scheduler to centralize deployment of internal APIs and integrated Prometheus with unified logging, reducing debugging time by 60% and streamlining future rollouts.

Data Analyst Intern at Mahindra and Mahindra Ltd. | Mumbai, India

January 2023 – April 2023

- Implemented a time-series forecasting model using ARIMAX and SARIMAX to predict financial trends for the next 6 years with 87% accuracy using past 20 years of monthly microeconomic data, which helped forecast sales volume across multiple car models.
- Compiled and processed a 30K+ record dataset across 20+ car models to analyze part cross-utilization, supporting cost optimization efforts.

Data Analyst Intern at Tech Mahindra Ltd. | Pune, India

June 2022 – August 2022

- Cleaned and transformed CRM data to improve usability and constructed interactive Tableau dashboards to deliver real-time visibility of customer deliverables, enabling improved status tracking and reporting accuracy.

PROJECTS

California Live Wildfire Dashboard

December 2025

- Architected a Calfire ETL pipeline using AWS Lambda and Amazon S3 for preprocessing and Google BigQuery for storage, processing 1000s of records per refresh by updating existing incidents and ingesting new ones for dashboard-ready analytics.
- Built a live Tableau wildfire analytics dashboard with YoY acres burned & YoY incident count KPIs, plus county and seasonal breakdowns, enabling faster identification of high-impact regions and trend shifts.

IndyGo – Ridership Dashboard (Client Project, ENGR-E 583: Information Visualization)

May 2025

- Ensured ridership accuracy by validating 38K GPS and APC records, eliminating 3% invalid entries through trip and route-specific thresholds.
- Designed an interactive Power BI dashboard analyzing 35K+ valid records, surfacing weekday vs weekend travel patterns, stop-level passenger flows, and route-level ridership KPIs, which led to a 3% improvement in service planning KPIs.

DocuVerse – ChatBot

January 2025

- Developed a RAG-based document processing app with LangChain and Hugging Face embeddings to automate text chunking and semantic representation, reducing manual document review time by 80%.
- Enhanced retrieval with FAISS and integrated Gemma2-9b-it to enable context-aware Q&A and structured summarization. Implemented persistent chat history and deployed the system on Streamlit for real-time usage.

CERTIFICATIONS

AWS Certified Cloud Practitioner (January 2026)